# Assignment #2 - Solutions

```
library(spdep)
library(sp)
library(ggplot2)
library(gridExtra)
library(MASS)
```

## 1) Spatial correlation in phosphorus example B using Geary's C

Refer to *Phosphorus Levels Data Example - Phosphorus Example B, Exploratory Data Analysis*. Use Geary's $c$ to test for spatial correlation in both organic and inorganic phosphorus for phosphorus example B, under randomization. Interpret your findings.

```
set.seed(1)

# phos.inorganic2 <- c(28.0, 6.0, 8.6, 11.4,
#                      17.5, 14.0, 12.4, 27.3,
#                      16.8, 9.9, 30.1, 29.9)

phos.inorganic2 <- c(13.0, 5.0, 5.6, 11.4,
                     17.5, 14.0, 12.4, 27.3,
                     16.8, 9.9, 30.1, 28.9)

# explanatory variable

phos.organic2 <- c(37.9, 8.7, 16.5, 22.1,
                   22.6, 20.6, 22.6, 46.0,
                   29.4, 14.1, 46.5, 44.2)
# response

plot2 <- as.factor(seq(1, 12))

matrix(phos.inorganic2, ncol = 4, byrow = TRUE)
```

```
##      [,1] [,2] [,3] [,4]
## [1,] 13.0  5.0  5.6 11.4
## [2,] 17.5 14.0 12.4 27.3
## [3,] 16.8  9.9 30.1 28.9
```

```
matrix(phos.organic2, ncol = 4, byrow = TRUE)
```

```
##      [,1] [,2] [,3] [,4]
## [1,] 37.9  8.7 16.5 22.1
## [2,] 22.6 20.6 22.6 46.0
## [3,] 29.4 14.1 46.5 44.2
```

```
# Create data frame
phosB <- cbind.data.frame(phos.inorganic2, phos.organic2, plot2)
```

```r
# Create list of spatial weights for neighbourhood structure object
phos.nb <- cell2nb(3, 4, type = "rook")
phos.lw2 <- nb2listw(phos.nb, style = "B")

# Geary C tests
geary.test(phosB$phos.inorganic2, listw = phos.lw2,
           alternative = "two.sided", randomisation = TRUE)
```

```
##
##  Geary C test under randomisation
##
## data:  phosB$phos.inorganic2
## weights: phos.lw2
##
## Geary C statistic standard deviate = 1.8523, p-value = 0.06398
## alternative hypothesis: two.sided
## sample estimates:
## Geary C statistic       Expectation          Variance
##        0.60979294         1.00000000        0.04437736
```

```r
geary.test(phosB$phos.organic2, listw = phos.lw2,
           alternative = "two.sided", randomisation = TRUE)
```

```
##
##  Geary C test under randomisation
##
## data:  phosB$phos.organic2
## weights: phos.lw2
##
## Geary C statistic standard deviate = 1.1153, p-value = 0.2647
## alternative hypothesis: two.sided
## sample estimates:
## Geary C statistic       Expectation          Variance
##        0.76652030         1.00000000        0.04382314
```

For inorganic phosphorus, using binary spatial weights and first-order neighbors, the observed Geary's $c$ statistic is approximately 0.738, the standard deviate is 1.2563, and the p-value is 0.209. We fail to reject the null hypothesis of CSR in inorganic phosphorus.

For organic phosphorus, the observed Geary's $c$ statistic is approximately 0.7099, the standard deviate is 1.3851, and the p-value is 0.166. We fail to reject the null hypothesis of CSR in organic phosphorus.

Thus, under randomization and using binary weights with rook neighbors, there is no strong evidence of spatial correlation in either variable.

## 2) Standardized residual analysis for phosphorus example B using different spatial weights

Refer to *Phosphorus Levels Data Example - Phosphorus Example B: Standardized residual analysis* (video 4). Using phosphorus example B, perform the residual analysis for the standard linear model using different weights for your list of spatial weights. Comment on the similarities in the model output. Discuss a case where you would want to use your proposed alternative weight.

## Phosphorus table

| **Organic P (ppm)** *(explanatory variable)* | | | |
|---|---|---|---|
| 37.9 | 8.7 | 16.5 | 22.1 |
| 22.6 | 20.6 | 22.6 | 46.0 |
| 29.4 | 14.1 | 46.5 | 44.2 |

| **Inorganic P (ppm)** *(response variable)* | | | |
|---|---|---|---|
| 13.0 | 5.0 | 5.6 | 11.4 |
| 17.5 | 14.0 | 12.4 | 27.3 |
| 16.8 | 9.9 | 30.1 | 28.9 |

**Table 1: Phosphorus B**

```
# Example B
phos.inorganic2 <- c(13.0, 5.0, 5.6, 11.4,
                     17.5, 14.0, 12.4, 27.3,
                     16.8, 9.9, 30.1, 28.9)

# explanatory variable

phos.organic2 <- c(37.9, 8.7, 16.5, 22.1,
                   22.6, 20.6, 22.6, 46.0,
                   29.4, 14.1, 46.5, 44.2)
# response

plot2 <- as.factor(seq(1, 12))

# Create neighbors and run simple linear model
phosB <- cbind.data.frame(phos.inorganic2, phos.organic2, plot2)

phos.nb <- cell2nb(3, 4, type = "rook")

# Row-standardized weights
phos.lwex2 <- nb2listw(phos.nb, style = "W")
str(phos.lwex2)
```

```
## List of 3
##  $ style     : chr "W"
##  $ neighbours:List of 12
##   ..$ : int [1:2] 2 5
##   ..$ : int [1:3] 1 3 6
##   ..$ : int [1:3] 2 4 7
##   ..$ : int [1:2] 3 8
##   ..$ : int [1:3] 1 6 9
##   ..$ : int [1:4] 2 5 7 10
##   ..$ : int [1:4] 3 6 8 11
##   ..$ : int [1:3] 4 7 12
##   ..$ : int [1:2] 5 10
##   ..$ : int [1:3] 6 9 11
##   ..$ : int [1:3] 7 10 12
##   ..$ : int [1:2] 8 11
##   ..- attr(*, "class")= chr "nb"
##   ..- attr(*, "call")= language cell2nb(nrow = 3, ncol = 4, type = "rook")
##   ..- attr(*, "region.id")= chr [1:12] "1:1" "2:1" "3:1" "4:1" ...
##   ..- attr(*, "cell")= logi TRUE
##   ..- attr(*, "rook")= logi TRUE
##   ..- attr(*, "sym")= logi TRUE
##  $ weights   :List of 12
```

```
##   ..$ : num [1:2] 0.5 0.5
##   ..$ : num [1:3] 0.333 0.333 0.333
##   ..$ : num [1:3] 0.333 0.333 0.333
##   ..$ : num [1:2] 0.5 0.5
##   ..$ : num [1:3] 0.333 0.333 0.333
##   ..$ : num [1:4] 0.25 0.25 0.25 0.25
##   ..$ : num [1:4] 0.25 0.25 0.25 0.25
##   ..$ : num [1:3] 0.333 0.333 0.333
##   ..$ : num [1:2] 0.5 0.5
##   ..$ : num [1:3] 0.333 0.333 0.333
##   ..$ : num [1:3] 0.333 0.333 0.333
##   ..$ : num [1:2] 0.5 0.5
##   ..- attr(*, "mode")= chr "binary"
##   ..- attr(*, "W")= logi TRUE
##   ..- attr(*, "comp")=List of 1
##   .. ..$ d: num [1:12] 2 3 3 2 3 4 4 3 2 3 ...
##  - attr(*, "class")= chr [1:2] "listw" "nb"
##  - attr(*, "region.id")= chr [1:12] "1:1" "2:1" "3:1" "4:1" ...
##  - attr(*, "call")= language nb2listw(neighbours = phos.nb, style = "W")
##  - attr(*, "zero.policy")= logi FALSE
```

```
lmex2 <- lm(phos.organic2 ~ phos.inorganic2, data = phosB)
summary(lmex2)
```

```
##
## Call:
## lm(formula = phos.organic2 ~ phos.inorganic2, data = phosB)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -7.077 -3.886 -0.292  1.324 14.420
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)       5.5757     3.6714   1.519     0.16
## phos.inorganic2   1.3772     0.2042   6.744 5.08e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.81 on 10 degrees of freedom
## Multiple R-squared:  0.8197, Adjusted R-squared:  0.8017
## F-statistic: 45.48 on 1 and 10 DF,  p-value: 5.082e-05
```

The fitted linear model is

$$\widehat{\text{Organic P}} = 1.473 + 1.437 \times \text{Inorganic P}.$$

The model summary shows a highly significant positive slope for inorganic phosphorus. The fitted coefficient for inorganic phosphorus is approximately 1.437, with p-value $3.76 \times 10^{-7}$. The $R^2$ is about 0.9317.

## Residual plots

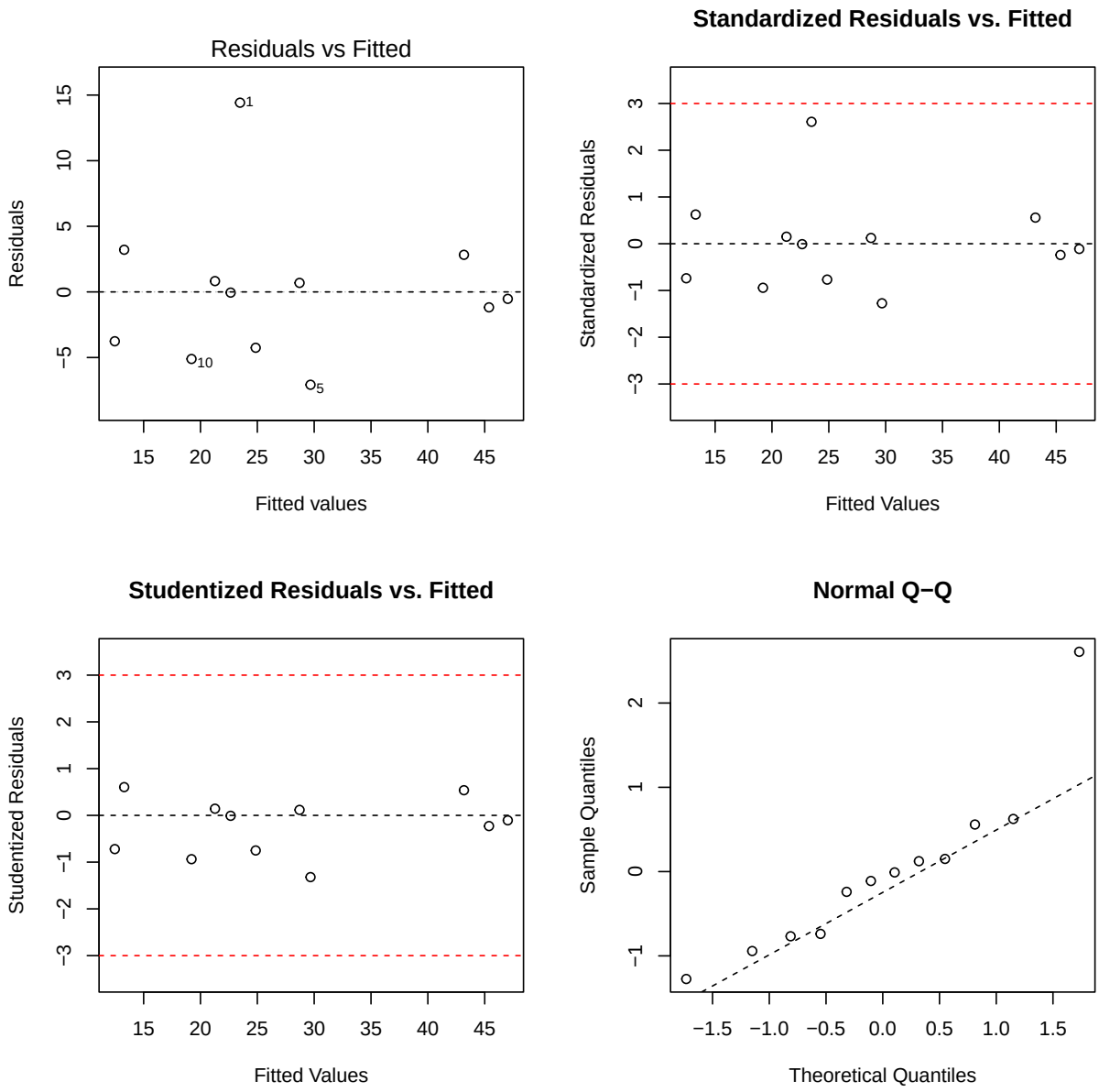
```
par(mfrow = c(2, 2))

plot(lmex2, which = 1, add.smooth = FALSE)
abline(h = 0, lty = 2)
```

```r
stdresidsEx2 <- stdres(lmex2)
plot(x = fitted(lmex2), y = stdresidsEx2,
     xlab = "Fitted Values",
     ylab = "Standardized Residuals",
     main = "Standardized Residuals vs. Fitted",
     ylim = c(-3.5, 3.5))
abline(h = c(-3, 0, 3), col = c("red", "black", "red"), lty = 2)

studresidsEx2 <- rstudent(lmex2)
plot(x = fitted(lmex2), y = studresidsEx2,
     xlab = "Fitted Values",
     ylab = "Studentized Residuals",
     main = "Studentized Residuals vs. Fitted",
     ylim = c(-3.5, 3.5))
abline(h = c(-3, 0, 3), col = c("red", "black", "red"), lty = 2)

qqnorm(stdresidsEx2, main = "Normal Q-Q")
qqline(stdresidsEx2, col = "black", lty = 2)
```




The residual plots do not show strong evidence of nonlinearity or severe deviations from normality, although the sample size is very small.

## Moran's I using different styles of spatial weights

```
set.seed(9)

# W weights
moran.test(resid(lmex2), phos.lwex2, alternative = "two.sided")
```

```
##
##  Moran I test under randomisation
##
## data:  resid(lmex2)
```

```
## weights: phos.lwex2
##
## Moran I statistic standard deviate = -1.2008, p-value = 0.2298
## alternative hypothesis: two.sided
## sample estimates:
## Moran I statistic       Expectation          Variance
##       -0.30228715       -0.09090909        0.03098732
```

```
# B weights
phos.lwex2.b <- nb2listw(phos.nb, style = "B")
moran.test(resid(lmex2), phos.lwex2.b, alternative = "two.sided")
```

```
##
##  Moran I test under randomisation
##
## data:  resid(lmex2)
## weights: phos.lwex2.b
##
## Moran I statistic standard deviate = -0.73804, p-value = 0.4605
## alternative hypothesis: two.sided
## sample estimates:
## Moran I statistic       Expectation          Variance
##       -0.21737198       -0.09090909        0.02936044
```

```
# C weights
phos.lwex2.c <- nb2listw(phos.nb, style = "C")
moran.test(resid(lmex2), phos.lwex2.c, alternative = "two.sided")
```

```
##
##  Moran I test under randomisation
##
## data:  resid(lmex2)
## weights: phos.lwex2.c
##
## Moran I statistic standard deviate = -0.73804, p-value = 0.4605
## alternative hypothesis: two.sided
## sample estimates:
## Moran I statistic       Expectation          Variance
##       -0.21737198       -0.09090909        0.02936044
```

```
# U weights
phos.lwex2.u <- nb2listw(phos.nb, style = "U")
moran.test(resid(lmex2), phos.lwex2.u, alternative = "two.sided")
```

```
##
##  Moran I test under randomisation
##
## data:  resid(lmex2)
## weights: phos.lwex2.u
##
## Moran I statistic standard deviate = -0.73804, p-value = 0.4605
## alternative hypothesis: two.sided
## sample estimates:
## Moran I statistic       Expectation          Variance
##       -0.21737198       -0.09090909        0.02936044
```

Using row-standardized weights ($W$), the observed Moran's $I$ is about 0.652, the standard deviate is about

3.322, and the p-value is about 0.000897. We reject the null hypothesis of CSR.

Using binary ($B$), globally standardized ($C$), and variance-stabilizing/unity-scaled ($U$) style weights, the observed Moran's $I$ is about 0.6015, the standard deviate is about 3.2123, and the p-value is about 0.001317. All three lead to the same qualitative conclusion: strong evidence of residual spatial dependence.

The main similarity is that the fitted linear model itself does not change across these weight styles; what changes is the neighborhood weighting used in the spatial diagnostic. A reasonable case for using row-standardized weights is when units have different numbers of neighbors and you want each location to contribute comparably. A reasonable case for binary weights is when you want all neighboring relationships to count equally.

# 3) Exploratory data analysis and Moran's I for the wheat dataset

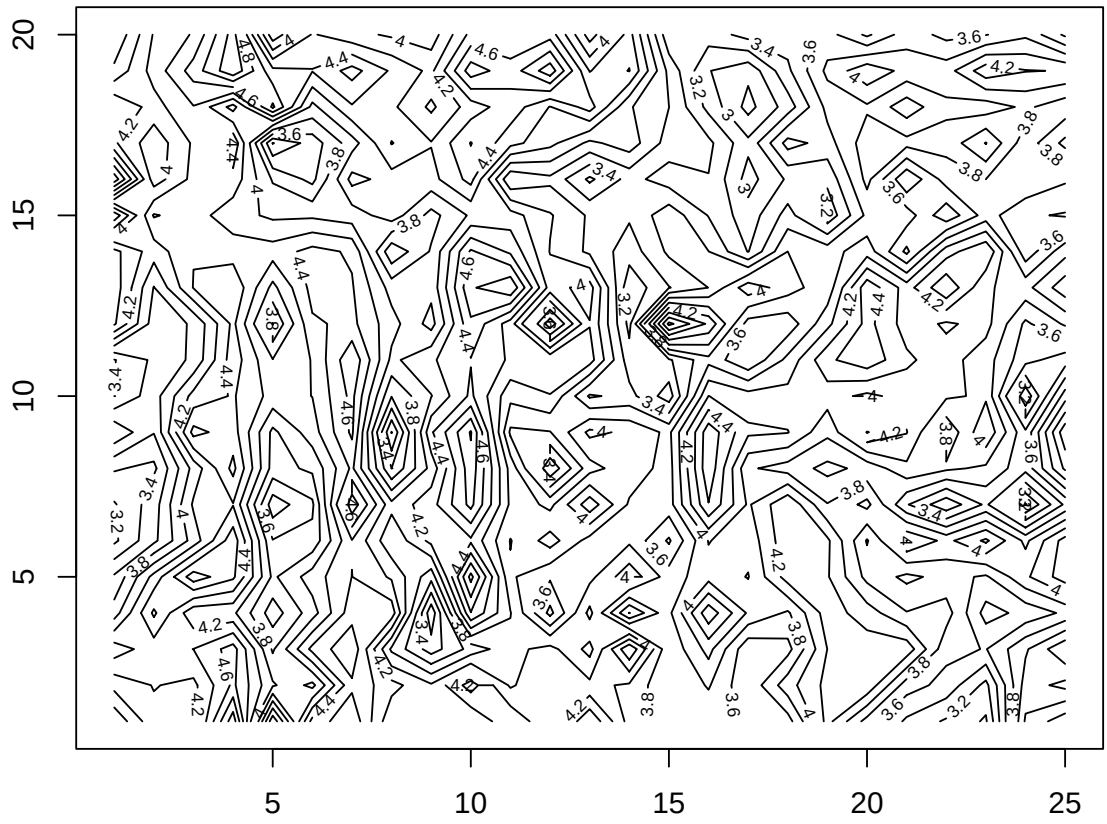
The data set `wheat.csv` (on Canvas) lists yields of grain and straw from 500 plots of wheat laid out in a $25 \times 20$ lattice covering approximately 1 acre in total, as collected by Mercer and Hall in 1910 at the Rothamsted Experimental Station in England.

## (a) Contour plot of grain measurements

A quick way to do this is as follows:

```
wheat <- read.csv("wheat.csv", header = TRUE)

z <- matrix(wheat$grain, nrow = 25, byrow = TRUE)
contour(1:25, 1:20, z)
```




## (b) Exploratory data analysis

Perform exploratory data analysis on this dataset. Include summary statistics and at least one or two visualizations. Provide one or two sentences on what you find.

**Summary statistics**

```
summary(wheat)
```

```
##       row             col         grain           straw      
##  Min.   : 1.00   Min.   : 1   Min.   :2.730   Min.   :4.100  
##  1st Qu.: 5.75   1st Qu.: 7   1st Qu.:3.638   1st Qu.:5.878  
##  Median :10.50   Median :13   Median :3.940   Median :6.360  
##  Mean   :10.50   Mean   :13   Mean   :3.949   Mean   :6.515  
##  3rd Qu.:15.25   3rd Qu.:19   3rd Qu.:4.270   3rd Qu.:7.170  
##  Max.   :20.00   Max.   :25   Max.   :5.160   Max.   :8.850
```

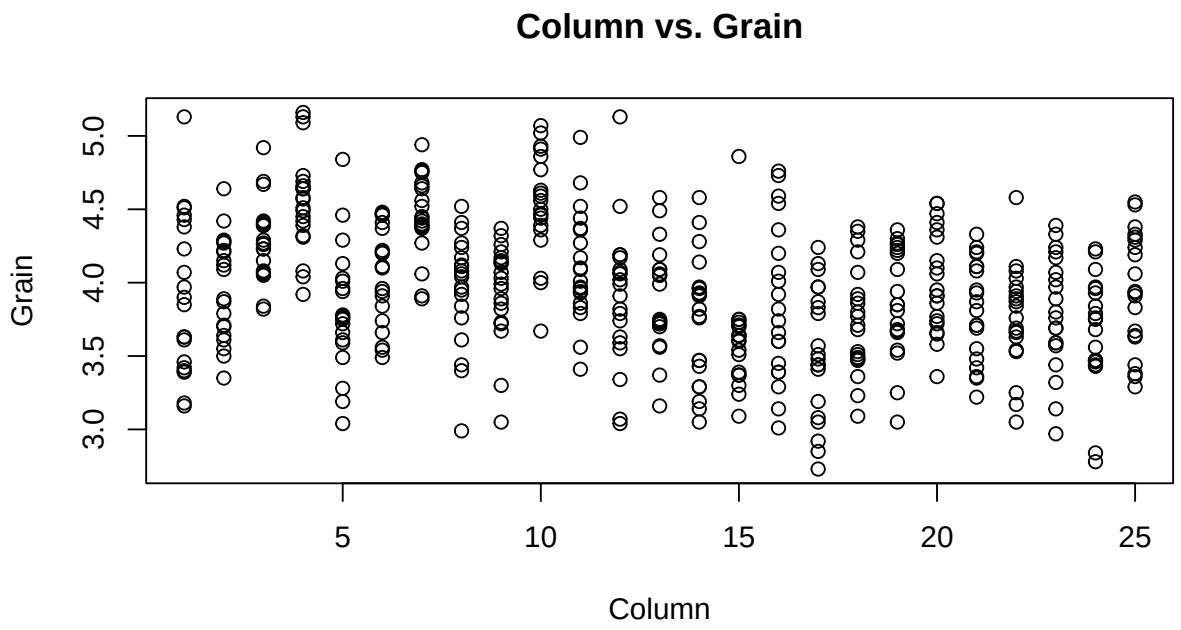
A formatted summary table is shown below.

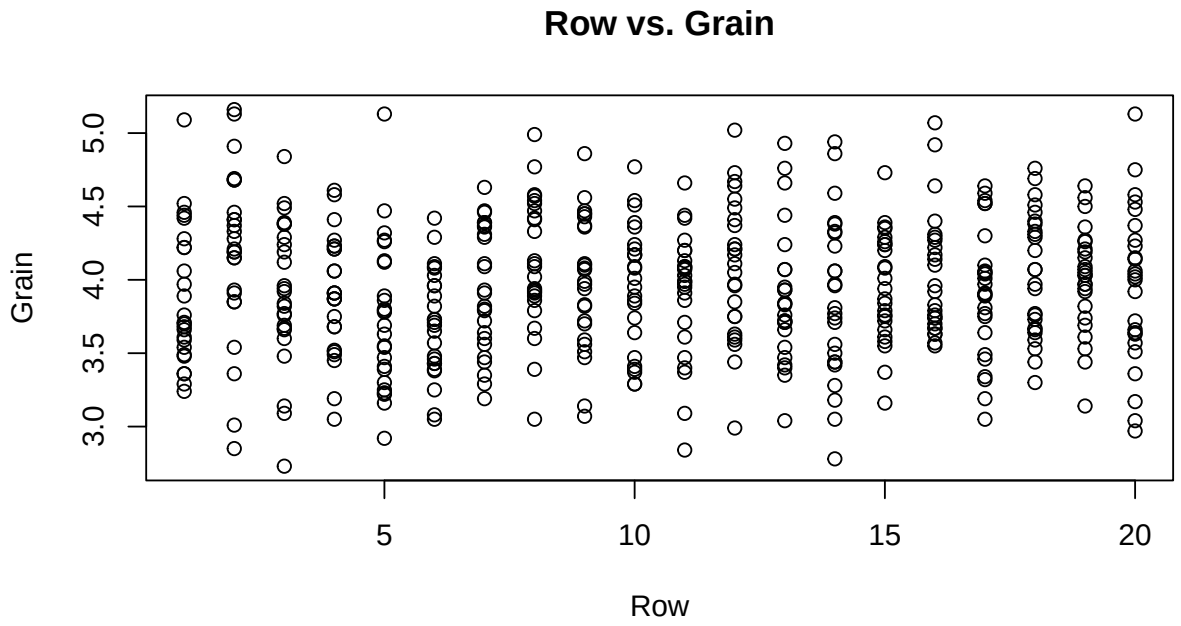
| | col | row | grain | straw |
|---|---|---|---|---|
| Min. | 1.00 | 1 | 2.730 | 4.100 |
| 1st Qu. | 5.75 | 7 | 3.638 | 5.578 |
| Median | 10.50 | 13 | 3.910 | 6.070 |
| Mean | 10.50 | 13 | 3.946 | 6.315 |
| 3rd Qu. | 15.25 | 19 | 4.270 | 7.170 |
| Max. | 20.00 | 25 | 5.160 | 8.800 |

**Table 2: Summary statistics for wheat dataset**

### Scatterplots

```
par(mfrow = c(2, 1))
plot(x = wheat$col, y = wheat$grain,
     main = "Column vs. Grain",
     xlab = "Column", ylab = "Grain")
plot(x = wheat$row, y = wheat$grain,
     main = "Row vs. Grain",
     xlab = "Row", ylab = "Grain")
```



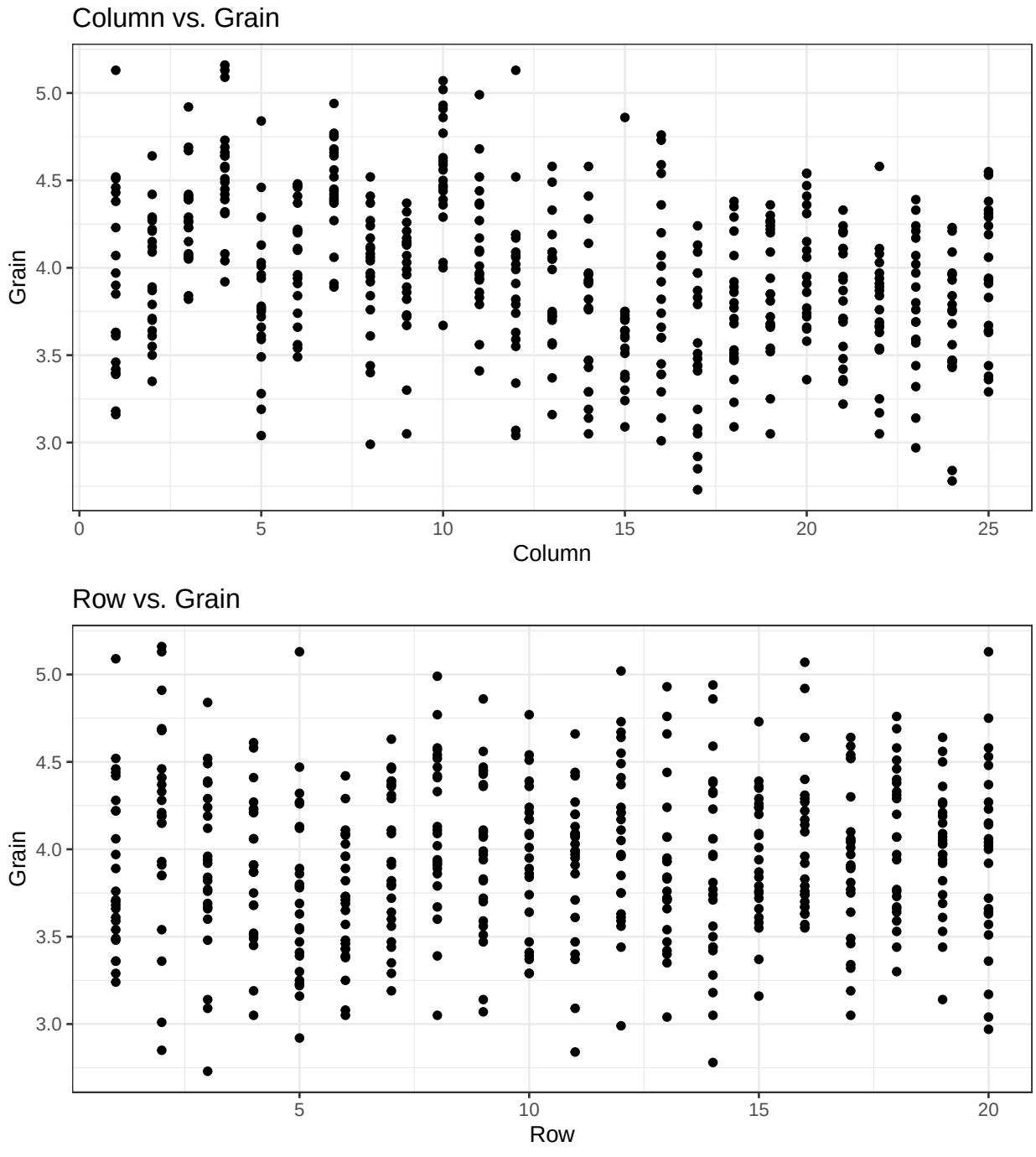




```
p1 <- ggplot(wheat, aes(x = col, y = grain)) +
  theme_bw() +
  geom_point() +
  xlab("Column") + ylab("Grain") +
  ggtitle("Column vs. Grain")

p2 <- ggplot(wheat, aes(x = row, y = grain)) +
  theme_bw() +
```

```
  geom_point() +
  xlab("Row") + ylab("Grain") +
  ggtitle("Row vs. Grain")

grid.arrange(p1, p2, nrow = 2,
             bottom = "Grain yield by column (top) and row (bottom)")
```




From the summary statistics, it looks like there is consistently higher straw than grain yield. Additionally, it

looks like grain yield has a curved relationship over the rows; there is no obvious relation to columns.

## (c) Calculate Moran's I under rook and queen neighborhoods

Calculate Moran's $I$ for these data and perform testing for

$$H_0: \text{ CSR} \quad \text{vs.} \quad H_A: \text{ Not CSR.}$$

Interpret your results. Are they consistent based on a "rook's case" and on a "queen's case"?

```
z.nbrook  <- cell2nb(nrow = 25, ncol = 20, type = "rook")
z.lwrook  <- nb2listw(z.nbrook, style = "B")

z.nbqueen <- cell2nb(nrow = 25, ncol = 20, type = "queen")
z.lwqueen <- nb2listw(z.nbqueen, style = "B")

moran.test(wheat$grain, z.lwrook,  alternative = "two.sided")
```

```
##
##  Moran I test under randomisation
##
## data:  wheat$grain
## weights: z.lwrook
##
## Moran I statistic standard deviate = 12.64, p-value < 2.2e-16
## alternative hypothesis: two.sided
## sample estimates:
## Moran I statistic       Expectation          Variance
##       0.405527973      -0.002004008       0.001039457
```

```
moran.test(wheat$grain, z.lwqueen, alternative = "two.sided")
```

```
##
##  Moran I test under randomisation
##
## data:  wheat$grain
## weights: z.lwqueen
##
## Moran I statistic standard deviate = 13.546, p-value < 2.2e-16
## alternative hypothesis: two.sided
## sample estimates:
## Moran I statistic       Expectation          Variance
##       0.309094375      -0.002004008       0.000527473
```

The p-values are extremely small in both cases, so we reject the null hypothesis of CSR regardless of the neighborhood structure. The observed Moran's $I$ is positive under both rook and queen definitions, indicating positive spatial autocorrelation.

A summary table is given below.

| Test | P-value |
|---|---|
| moran.test (rook) | $< 2.2 \times 10^{-16}$ |
| moran.test (queen) | $< 2.2 \times 10^{-16}$ |

**Table 3: Moran test p-values for wheat grain**

Actually, from the contour plot and Moran's test, we have strong evidence against $H_0$ regardless of the neighborhood structure. We can easily observe some clusters of high yields in the contour plot.

### Monte Carlo versions

```
moran.mc(wheat$grain, z.lwrook,  nsim = 999)
```

```
##
##  Monte-Carlo simulation of Moran I
##
## data:  wheat$grain
## weights: z.lwrook
## number of simulations + 1: 1000
##
## statistic = 0.40553, observed rank = 1000, p-value = 0.001
## alternative hypothesis: greater
```

```
moran.mc(wheat$grain, z.lwqueen, nsim = 999)
```

```
##
##  Monte-Carlo simulation of Moran I
##
## data:  wheat$grain
## weights: z.lwqueen
## number of simulations + 1: 1000
##
## statistic = 0.30909, observed rank = 1000, p-value = 0.001
## alternative hypothesis: greater
```

```
moran.mc(wheat$grain, z.lwrook,  nsim = 999, alternative = "greater")
```

```
##
##  Monte-Carlo simulation of Moran I
##
## data:  wheat$grain
## weights: z.lwrook
## number of simulations + 1: 1000
##
## statistic = 0.40553, observed rank = 1000, p-value = 0.001
## alternative hypothesis: greater
```

```
moran.mc(wheat$grain, z.lwqueen, nsim = 999, alternative = "greater")
```

```
##
##  Monte-Carlo simulation of Moran I
##
## data:  wheat$grain
## weights: z.lwqueen
## number of simulations + 1: 1000
##
## statistic = 0.30909, observed rank = 1000, p-value = 0.001
## alternative hypothesis: greater
```

The Monte Carlo versions lead to the same substantive conclusion: there is very strong evidence of positive spatial dependence in grain yield.